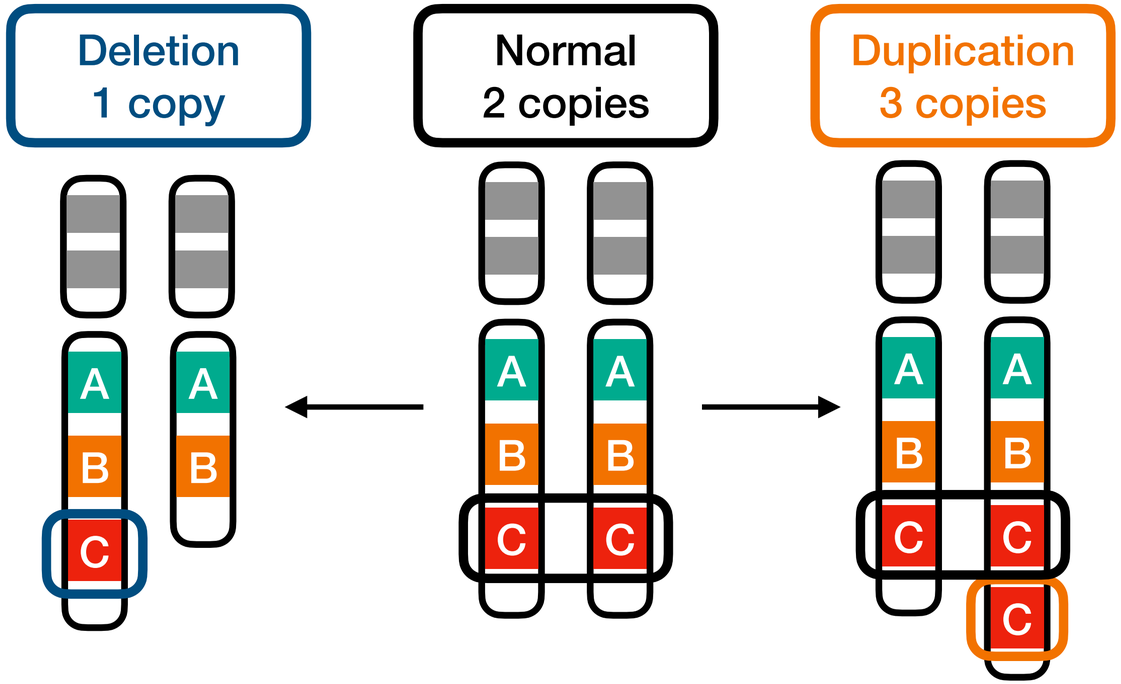
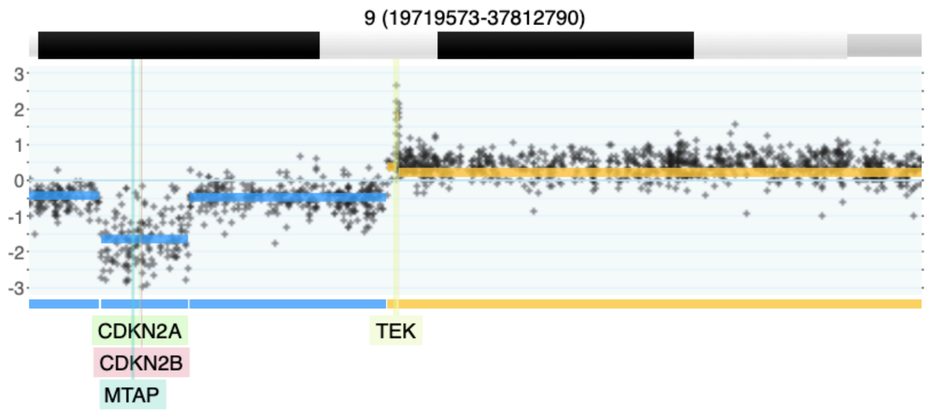
Introduction to Progenetix and Cancer Classification

Jiahui Yu / 16-04-2026

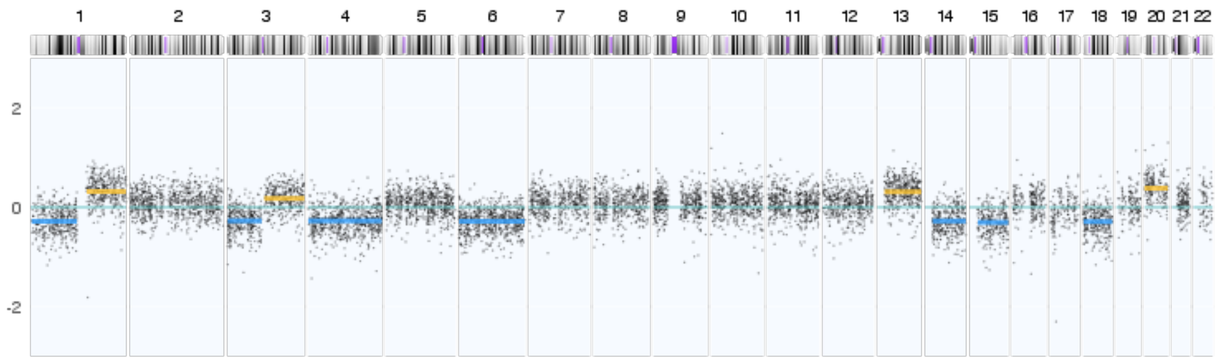
Copy Number Variant (CNV)



- Intermediate-scale genetic change
- Size: 1kb to multiple megabase
- Additional copies of sequence (duplications) and losses of genetic material (deletions)



2-event, homozygous deletion in a Glioblastoma

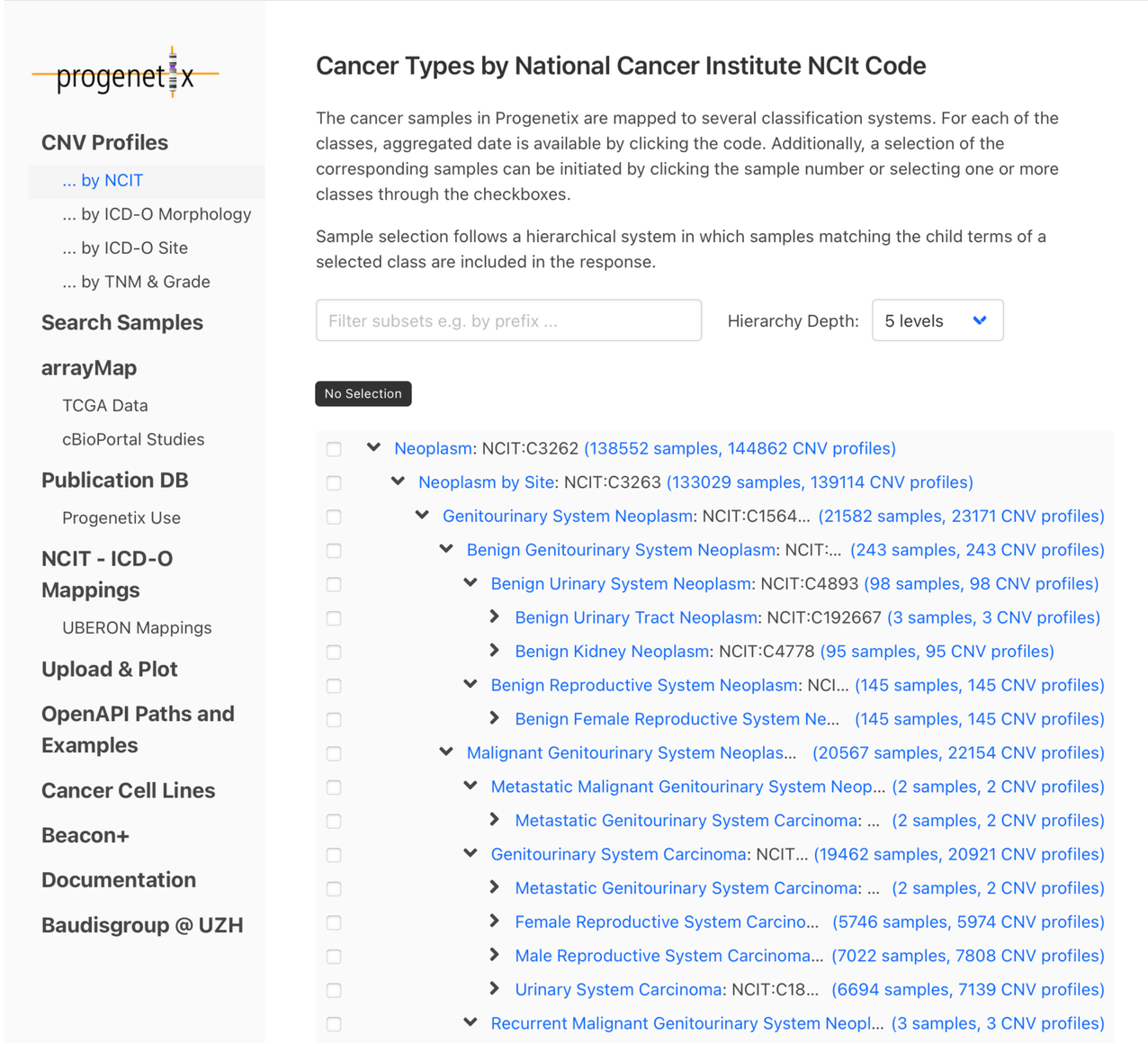


Gain of chromosome arm 13q in colorectal carcinoma

progenetix.org

Cancer Genomics Reference Resource

- ***open*** resource for oncogenomic profiles
- over **240'000 cancer CNV profiles**
- more than **1100 diagnostic types**
- runs on a **Beacon API**
- inclusion of reference datasets (e.g. TCGA)
- support for SNV data (TCGA, cell lines...)
- standardized encodings (e.g. NCIt, ICD-O 3)
- identifier mapping for PMID, GEO, Cellosaurus, TCGA, cBioPortal where appropriate
- core clinical data (TNM, sex, survival ...)
- data mapping services



Universität Zürich[UZH]

progenetix

SIB Swiss Institute of Bioinformatics

How are cancer sample labels in Progenetix

- NCIt: hierarchical disease classification
- ICD-O Morphology: tumor type/morphology
- ICD-O Topography: anatomical site
- **NCIt** (NCI Thesaurus) combines disease concepts into one hierarchical label
- **ICD-O** (International Classification of Diseases for Oncology) separates morphology and anatomical site into two coding axes
- Example
  - pgx:icdom-81403: Adenocarcinoma, NOS
  - pgx:icdot-C56.9: Ovary
  - NCIT:C7700: Ovarian adenocarcinoma

What kind of data do we get?

- **Sample data/metadata (Biosamples)**: one row per biosample
  - identifiers, diagnosis, sex, age, tissue, sample status, etc.

Results | Biosamples | Variants

Biosample Id	Dx Classifications	Identifiers
pgxbs-m84dh8s2	pgx:icdom-94403 Glioblastoma, NOS pgx:icdot-C71.9 Brain, NOS NCIT:C3058 Glioblastoma	
pgxbs-m84cz2cc	pgx:icdom-94403 Glioblastoma, NOS pgx:icdot-C71.9 Brain, NOS NCIT:C3058 Glioblastoma	
pgxbs-m84dd1zf	pgx:icdom-94403 Glioblastoma, NOS pgx:icdot-C71.9 Brain, NOS NCIT:C3058 Glioblastoma	
pgxbs-kftvil7b	pgx:icdom-94403 Glioblastoma, NOS pgx:icdot-C71.9 Brain, NOS NCIT:C3058 Glioblastoma	pubmed:18215105 Cloughesy TF, Yoshimoto K et al. (2008): Antitumor activity of rapamycin in a...
pgxbs-m84d05e1	pgx:icdom-94403 Glioblastoma, NOS pgx:icdot-C71.9 Brain, NOS NCIT:C3058 Glioblastoma	
pgxbs-kftvku8q	pgx:icdom-94403 Glioblastoma, NOS pgx:icdot-C71.9 Brain, NOS NCIT:C3058 Glioblastoma	pubmed:28481359 Zehir A, Benayed R et al. (2017): Mutational landscape of metastatic cancer revealed... cbioportal:msk_impact_2017 Targeted sequencing of 10,000 clinical cases using the MSK-IMPACT assay
pgxbs-m84d328c	pgx:icdom-94403 Glioblastoma, NOS pgx:icdot-C71.9 Brain, NOS NCIT:C3058 Glioblastoma	
pgxbs-kftvib0z	pgx:icdom-94403 Glioblastoma, NOS pgx:icdot-C71.9 Brain, NOS NCIT:C3058 Glioblastoma	pubmed:19074898 Solomon DA, Kim JS et al. (2008): Mutational inactivation of PTPRD in glioblastoma...
pgxbs-m84dguzb	pgx:icdom-94403 Glioblastoma, NOS pgx:icdot-C71.9 Brain, NOS NCIT:C3058 Glioblastoma	
pgxbs-kftvkuac	pgx:icdom-94403 Glioblastoma, NOS pgx:icdot-C71.9 Brain, NOS NCIT:C3058 Glioblastoma	pubmed:28481359 Zehir A, Benayed R et al. (2017): Mutational landscape of metastatic cancer revealed... cbioportal:msk_impact_2017 Targeted sequencing of 10,000 clinical cases using the MSK-IMPACT assay

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What kind of data do we get?

- **Variants data:** one row per CNV segment
  - chromosome, start, end, CNV state

Results | Biosamples | Variants

Digest	Gene	Pathogenicity	Variant type	Variant Instances
NC_000013.11:48957415-48970526:EFO_0020073			CopyNumberChange	V: bycvar-6657d1aa02cc20f830464b3a A: pgxcs-kftwy8s2 B: pgxbs-kftvkl70 I: pgxind-kftx6pk5
NC_000001.11:77404639-77432003:EFO_0020073			CopyNumberChange	V: bycvar-66582ac902cc20f8304c8969 A: pgxcs-kftx0dya B: pgxbs-kftvkokk I: pgxind-kftx6ts8
NC_000005.10:102835209-108672054:EFO_0030068			CopyNumberChange	V: bycvar-6657cb266ac278bc0e53cf82 A: pgxcs-kftwg76p B: pgxbs-kftvj8w0 I: pgxind-kftx537c
NC_000004.12:182652615-190106768:EFO_0030068			CopyNumberChange	V: bycvar-66e303cfd2ef81e071d26835 A: pgxcs-kftwtz7d B: pgxbs-kftvksf6 I: pgxind-kftx6y7t
NC_000001.11:78243037-78282082:EFO_0030067			CopyNumberChange	V: bycvar-5bab57a2727983b2e00d5552 A: pgxcs-kftvmnll B: pgxbs-kftvgkdo I: pgxind-kftx2asm
NC_000017.11:43934327-45395141:EFO_0030071			CopyNumberChange	V: bycvar-665725b9183fc78d1fb12e88 A: pgxcs-kftw1tbd B: pgxbs-kftvhxyr I: pgxind-kftx3tkh
NC_000014.9:71698491-99196164:EFO_0030068			CopyNumberChange	V: bycvar-665726b18f6b961582619a29 A: pgxcs-kftw3yi5 B: pgxbs-kftvi4nj I: pgxind-kftx3vzk
NC_000008.11:11241951-11250222:EFO_0030070			CopyNumberChange	V: bycvar-5bab57b6727983b2e00dfbc1 A: pgxcs-kftvmor0 B: pgxbs-kftvgkh1 I: pgxind-kftx2awp
NC_000023.11:41456779-41579686:EFO_0030071			CopyNumberChange	V: bycvar-6656ffcbea1558707a981155 A: pgxcs-kftvz2p3 B: pgxbs-kftvhp46 I: pgxind-kftx3hdk
NC_000017.11:45382635-46062635:EFO_0030071			CopyNumberChange	V: bycvar-673cf6922cbf8b196b9d26a7 A: pgxcs-m3io6eby B: pgxbs-m3io6eby I: pgxind-m3io2hwq

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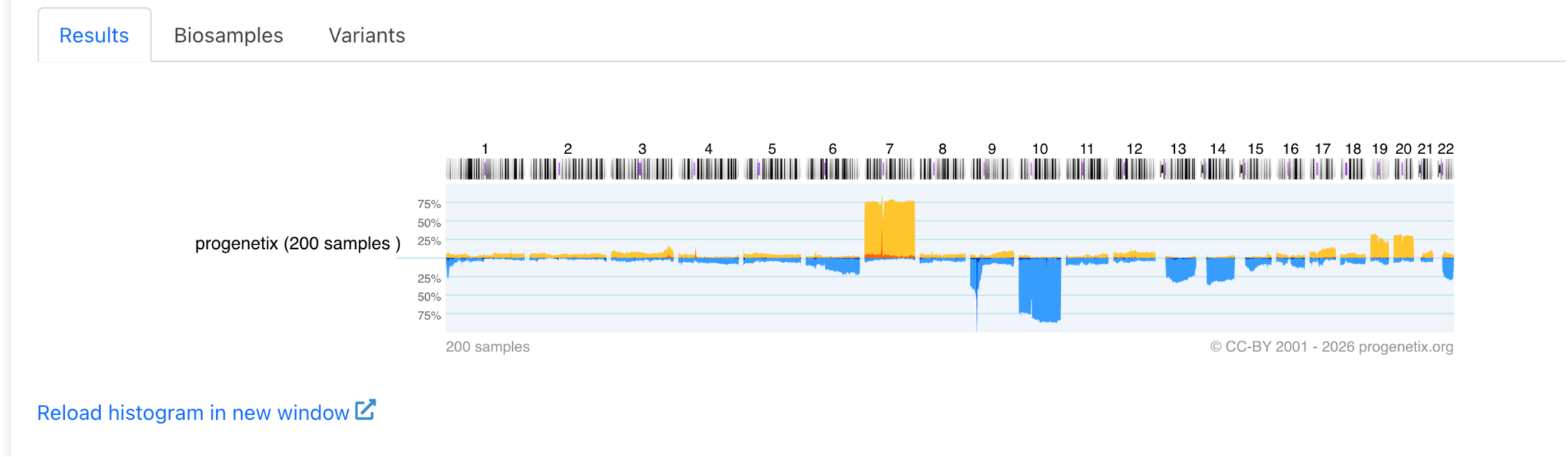
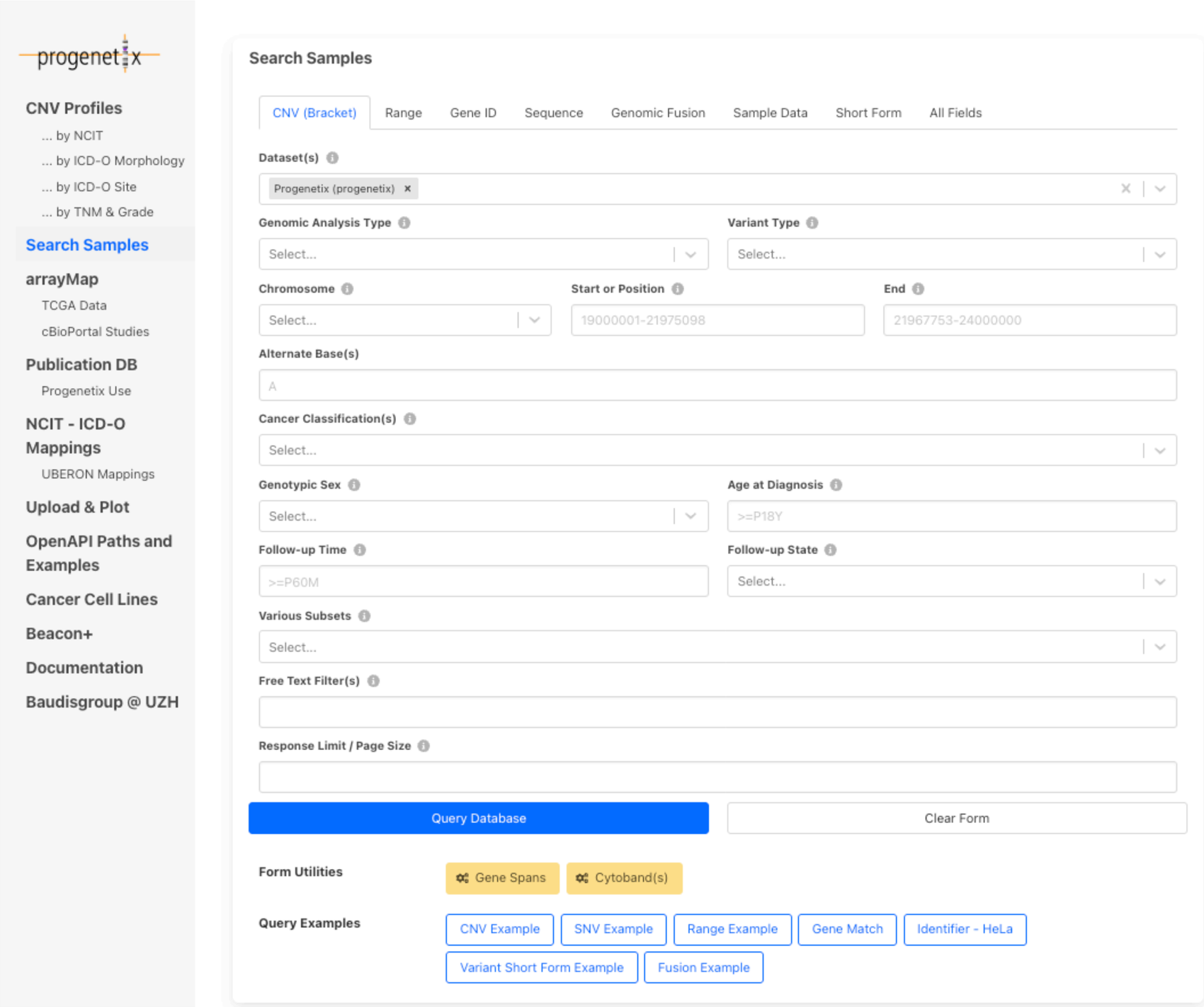
What kind of data do we get?

- **Frequency data**: group-level CNV summaries across many samples
  - default **1 Mb binned** representation



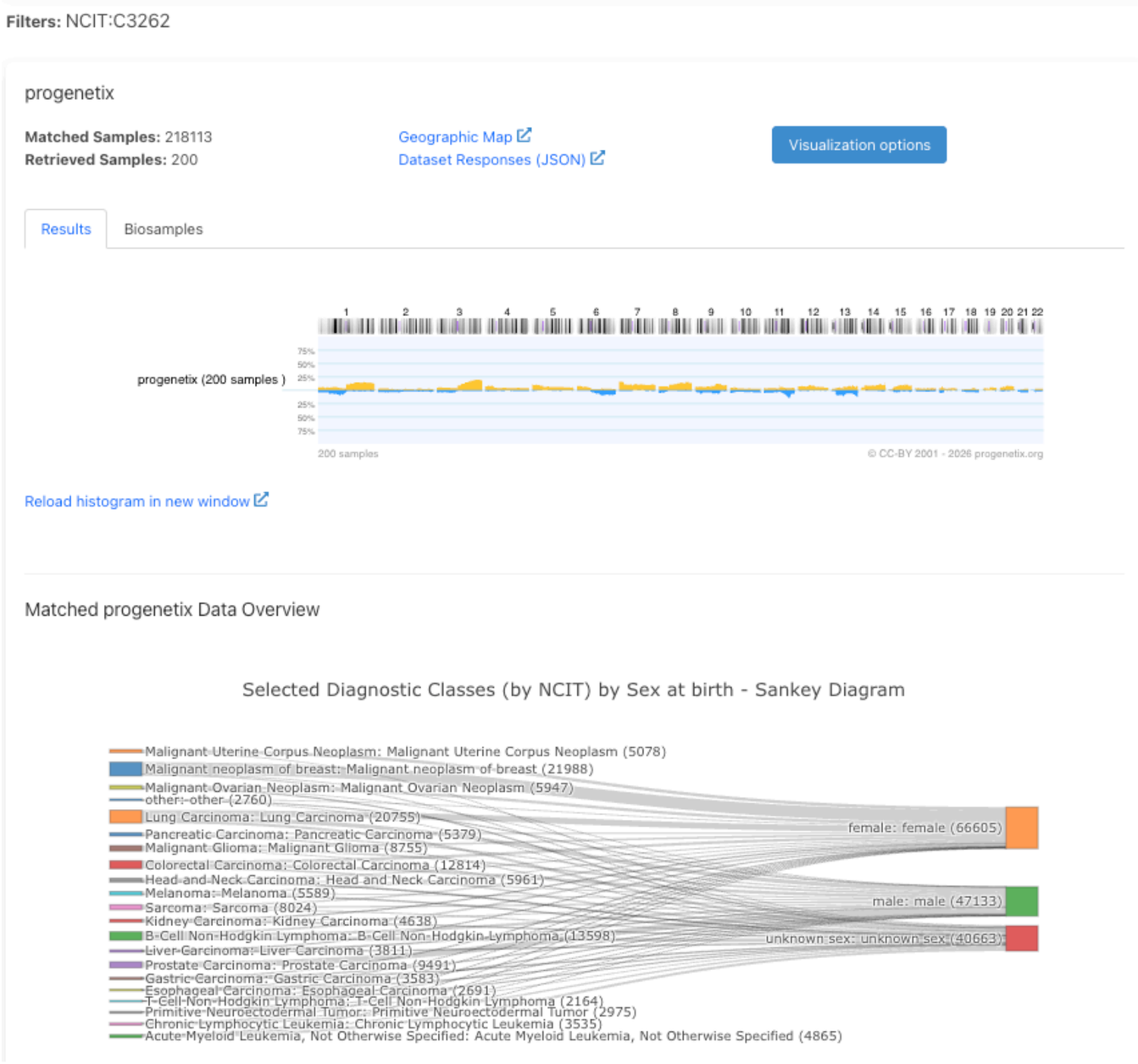
Explore Progenetix Webpage

progenetix.org



Small Task

- **Task 1**
- Find One cancer type of your choice in Progenetix
  - Write down:
    - Its classification label
    - The number of matched samples or profiles
    - One broader parent category, if shown
- **Task2**
- Search for one gene, for example, CDKN2A
- Check what kind of CNV-related query result is shown



Explore Progenetix Dataset

- Download and Go through the notebook below:
- https://github.com/compbiozurich/UZH-BIO392-FS26/blob/master/course-material/2026-04-16_day3_exercise/intro_progenetix.ipynb
- Finish exercises.

Read this paper and think about the following questions:

1. Why is important to construct a CNV map on health individuals of various ethnicities? **(Introduction)**
2. What is the CNV size that the authors defined? (Box 1 mentioned in **introduction**)

Noted: The CNV size definition is still under debate and may be different in other literatures

3. What are the primary approaches used for CNV detection? And what are the advantages and limitations of these technologies? **(CNV discoveries)**
4. The authors used clustering method to combine data from different studies into merged CNVRs (Copy number variable regions). What are the two criteria for cluster filtering? And why did they do this filtering? **(The CNV map)**
5. What are thresholds in stringency level 1, inclusive map (stringency level 2), and stringent map (stringency level 12) respectively? **(The CNV map)**
6. Which percentage of the genome contributes to CNV in inclusive map and stringent map respectively? (**Properties of the CNV map**)
7. By your intuition, which kind of genes are more variable between protein-coding genes and non-coding genes? How about their findings in this paper? **(Functional impact of CNV)**
8. The authors generated a null CNV map and found genes for which at least 85% of the exons were homozygous deleted. What are the functions of these genes? And why did the authors say that they seem to be non-essential? **(Homozygous deleted genes)**
9. If you are a medical doctor, how do you use this map as a tool to assess the clinical importance of a CNV? **(Clinical application of the CNV map part in Discussion)**

nature > nature reviews genetics > analyses > article

Published: 03 February 2015

A copy number variation map of the human genome

Mehdi Zarrei, Jeffrey R. MacDonald, Daniele Merico & Stephen W. Scherer

Nature Reviews Genetics **16**, 172–183 (2015) | Cite this article

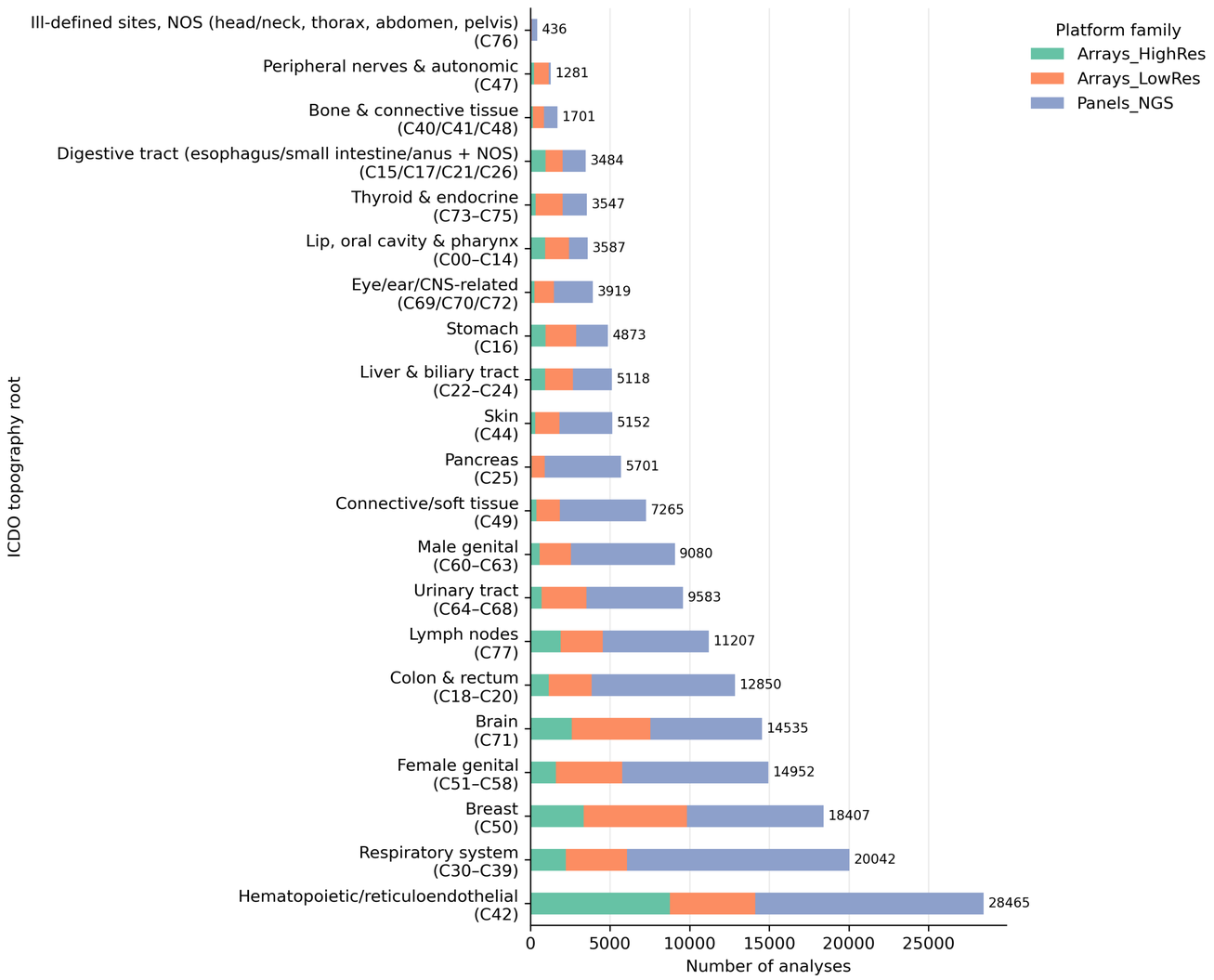
29k Accesses | **400** Citations | **63** Altmetric | Metrics

Paper link: https://www.nature.com/articles/nrg3871

Why data-driven cancer classfication

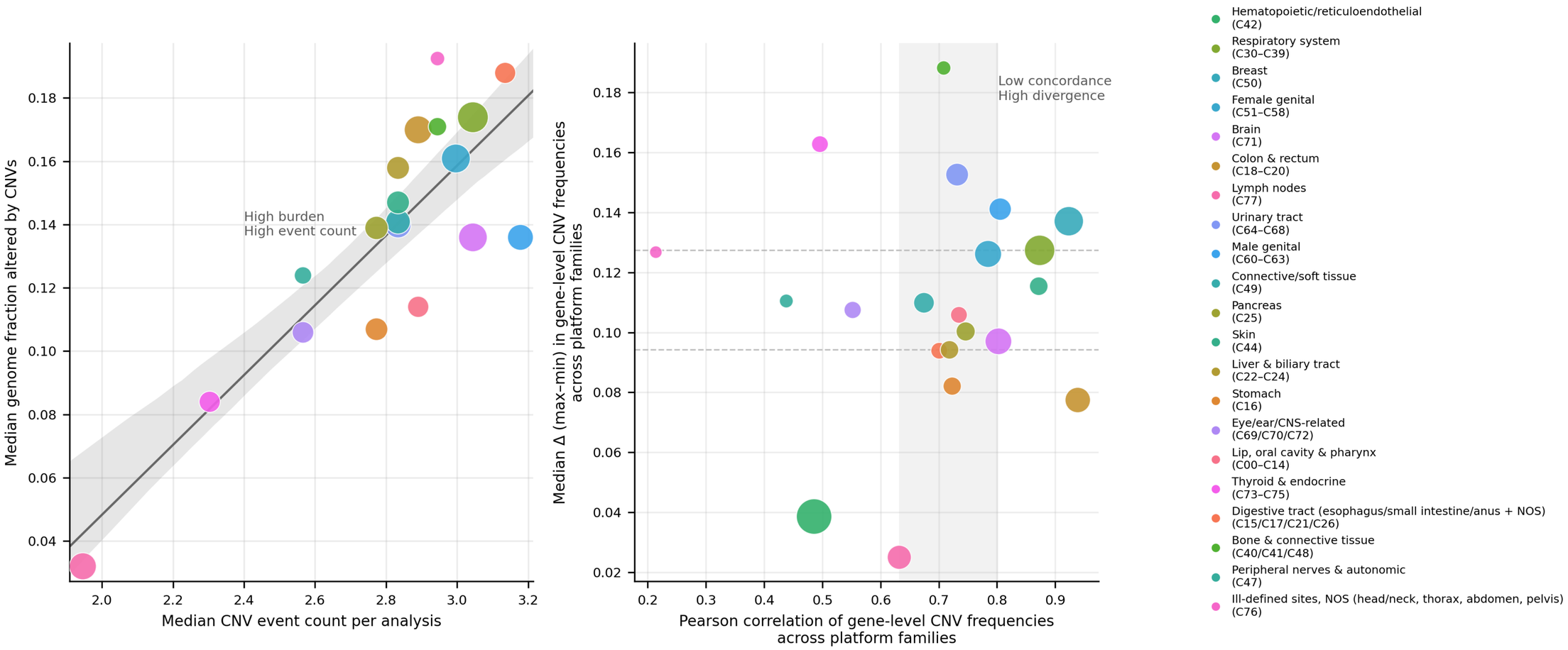
- Cancer classification systems such as NCIT and ICD-O are essential for annotation, communication, and data organization.
- However, these systems were not designed specifically as molecular feature spaces.
- A single label may still contain substantial biological heterogeneity.
- Different classification systems also emphasize different aspects:
  - disease ontology/hierarchy
  - anatomical site
  - morphology/histology
- Therefore, it is reasonable to ask whether existing labels are well supported by genomic data, and whether some groups are easier or harder to distinguish at the molecular level.

Real World Data



- The number of analyses differs across cancer groups.
- Platform families are unevenly represented across roots.
- Classification results can therefore reflect both biology and dataset structure.
- This motivates careful sampling and preprocessing.

Biologically and technically heterogenenous



What is data-driven classification

An ideal data-driven classifier is:

- **Biologically grounded**: distinguishes tumors because of meaningful molecular differences
- **Technically robust**: not dominated by platform effects or batch structure
- **Interpretable**: we can identify which features drive the separation
- **Generalizable**: works beyond the current sample set

Clinical Impact:

- distinguish groups with similar morphology but different genomic patterns
- identify ambiguous or borderline cases (cancer of unknown primary)
- reveal biologically meaningful subgroups
- support future biomarker development (cell lines)

How classification is usually approached

- **Supervised:**
  - We already have labels.
  - The goal is to learn how measured features relate to those labels.
  - Example: distinguishing two known tumor groups.
- **Unsupervised:**
  - We do not begin with fixed labels.
  - The goal is to detect structure, clusters, or subgroups directly from the data.
  - Example: asking whether samples form unexpected molecular subtypes.

Typical Workflow

- **Define the biological question**
  - Example: Do two tumor entities differ sufficiently in CNV architecture to support sample-level discrimination?
- **Define the units and groups**
  - Example: Treat each biosample as a single observation and assign labels based on predefined diagnostic categories.
- **Select measurable features**
  - Example: Represent each sample by genome-wide CNV bins or gene-level copy-number states.
- **Explore separability in the data**
  - Example: Examine whether labeled groups show distinct distributions, clusters, or burden patterns before formal modeling.
- **Build and evaluate a classification strategy**
  - Example: Fit a supervised model and evaluate its robustness using held-out data or repeated cross-validation.
- **Interpret the result biologically and clinically**
  - Example: Relate discriminative features to known cancer biology and consider whether they could support refinement of existing diagnostic groupings.

Classfication Showcase

- Download and Go through the notebook below:
- https://github.com/compbiozurich/UZH-BIO392-FS26/blob/master/course-material/2026-04-16_day3_exercise/intro_classification.ipynb
- Finish exercises.